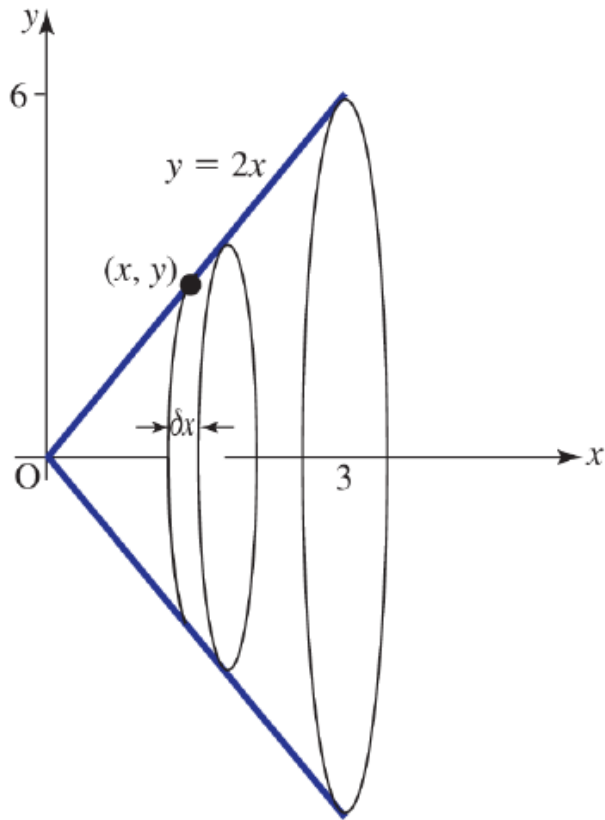


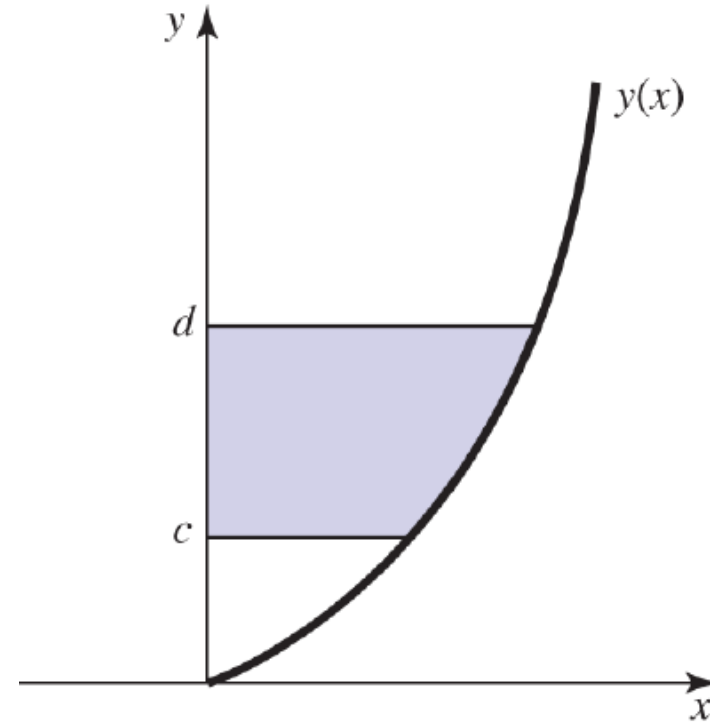
Revision L6 – L10

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L6. Volumes Generated by Rotating Areas about Axes

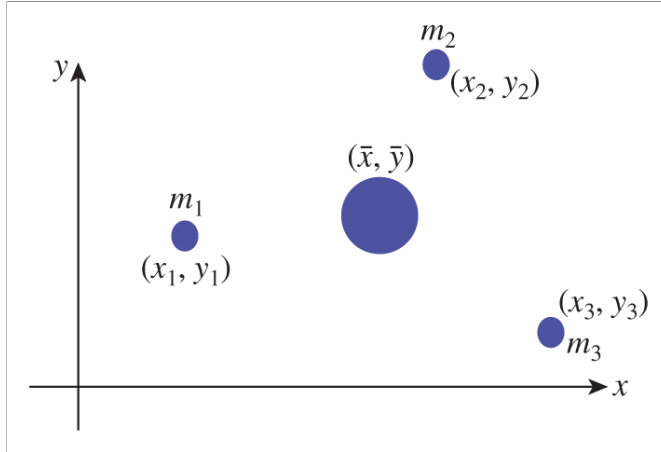


$$\lim_{\delta x \rightarrow 0} \sum_{x=a}^{x=b} \pi y^2 \delta x = \int_a^b \pi y^2 dx$$



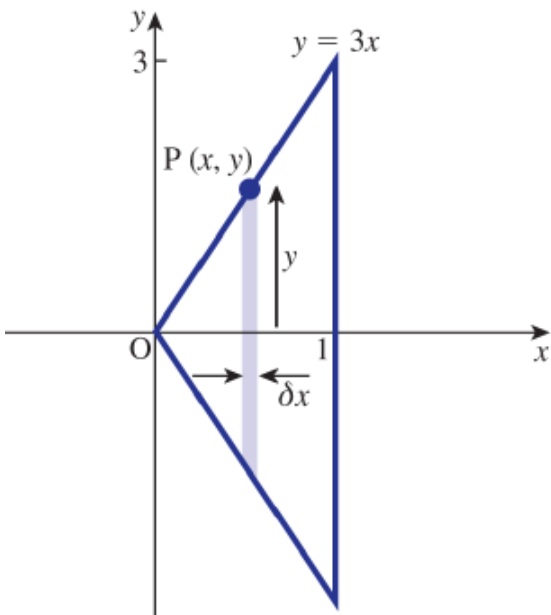
$$\lim_{\delta y \rightarrow 0} \sum_{y=c}^{y=d} \pi x^2 \delta y = \int_c^d \pi x^2 dy$$

L6. Calculation of Centre of Mass of a Lamina



$$\bar{x} = \frac{\sum_{i=1}^n m_i x_i}{\sum_{i=1}^n m_i},$$

$$\bar{y} = \frac{\sum_{i=1}^n m_i y_i}{\sum_{i=1}^n m_i}$$



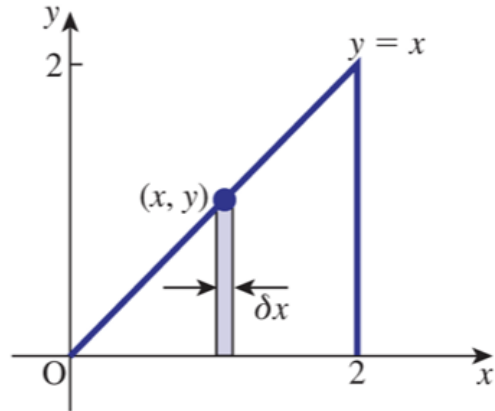
By symmetry the centre of mass lies on the x axis. Thus, $\bar{y} = 0$. To find $\bar{x}$:

- ① Divide the lamina into small pieces: thin vertical slices in this case
- ② Find the mass of each piece: $m(2y\delta x)$;
- ③ Calculate the moment of each piece about the y axis: $mx(2y\delta x)$;
- ④ The sum of these moments about the y axis will be equal to $M\bar{x}$:

$$\int_0^1 2mxy \, dx = M\bar{x}, \quad \text{with } M = 3m$$

L6. Calculation of Centre of Mass of a Lamina

5. Find the centre of mass of the uniform plane lamina below.



1) Total moment

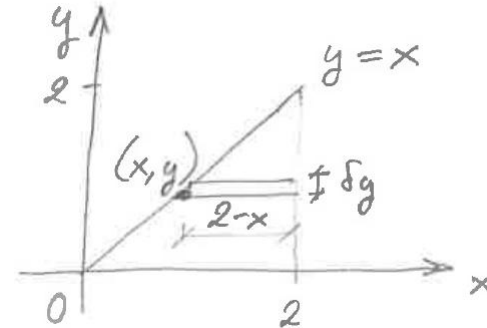
$$M1 = 2m\bar{x} \text{ (Area = 2, mass per unit area = m)}$$

2) Sum of moments of small strips

$$M2 = \sum_{x=0}^2 m(y\delta x)x = \sum_{x=0}^2 mx^2\delta x \rightarrow \int_0^2 mx^2 dx = \frac{8}{3}m$$

3) $M1 = M2$

$$2m\bar{x} = \frac{8}{3}m \rightarrow \bar{x} = \frac{4}{3}$$



1) Total moment

$$M1 = 2m\bar{y} \text{ (Area = 2, mass per unit area = m)}$$

2) Sum of moments of small strips

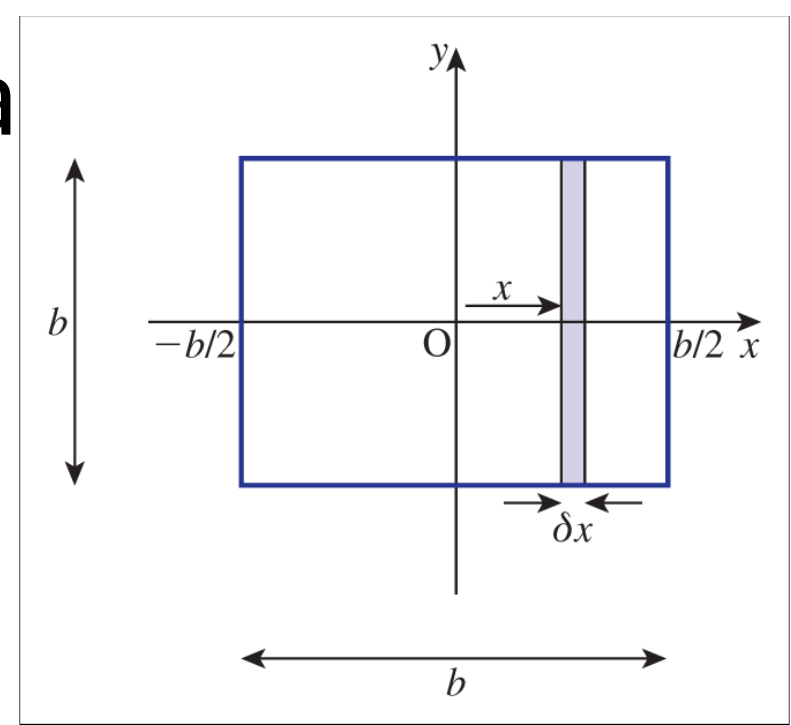
$$M2 = \sum_{x=0}^2 m((2-x)\delta y)y \rightarrow \int_0^2 m(2-y)y dy = \frac{4}{3}m$$

3) $M1 = M2$

$$2m\bar{y} = \frac{4}{3}m \rightarrow \bar{y} = \frac{2}{3}$$

L6. Calculation of Moment of Inertia

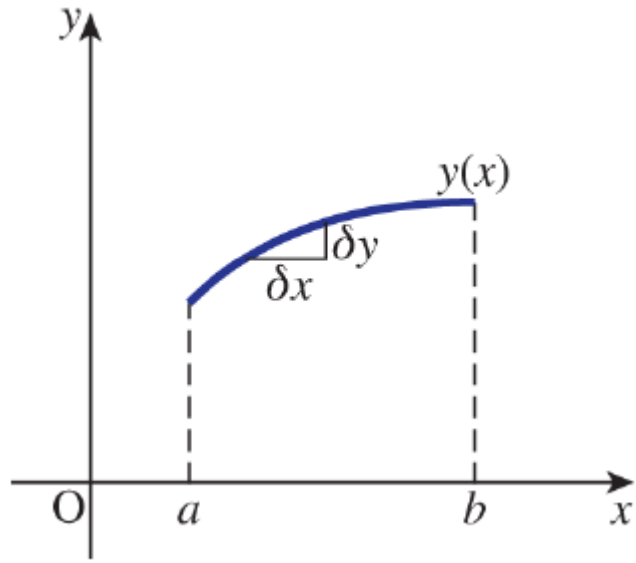
moment of inertia $I = \int r^2 dm$



- 1 Divide the lamina into many small pieces: thin vertical strips in this case;
- 2 Calculate the moment of inertia of each strip: $x^2 mb \delta x$;
- 3 Add contributions from all strips: $\sum_{x=-b/2}^{x=b/2} x^2 mb \delta x$;
- 4 Calculate I as the sum as $\delta x \rightarrow 0$: $I = \int_{-b/2}^{b/2} x^2 mb dx = \frac{mb^4}{12}$;
- 5 Express I in terms of the total mass M : $M = mb^2$, $I = \frac{Mb^2}{12}$.

Important: The limits on the integral must be chosen so that the whole lamina is included.

L6. Length of a Curve



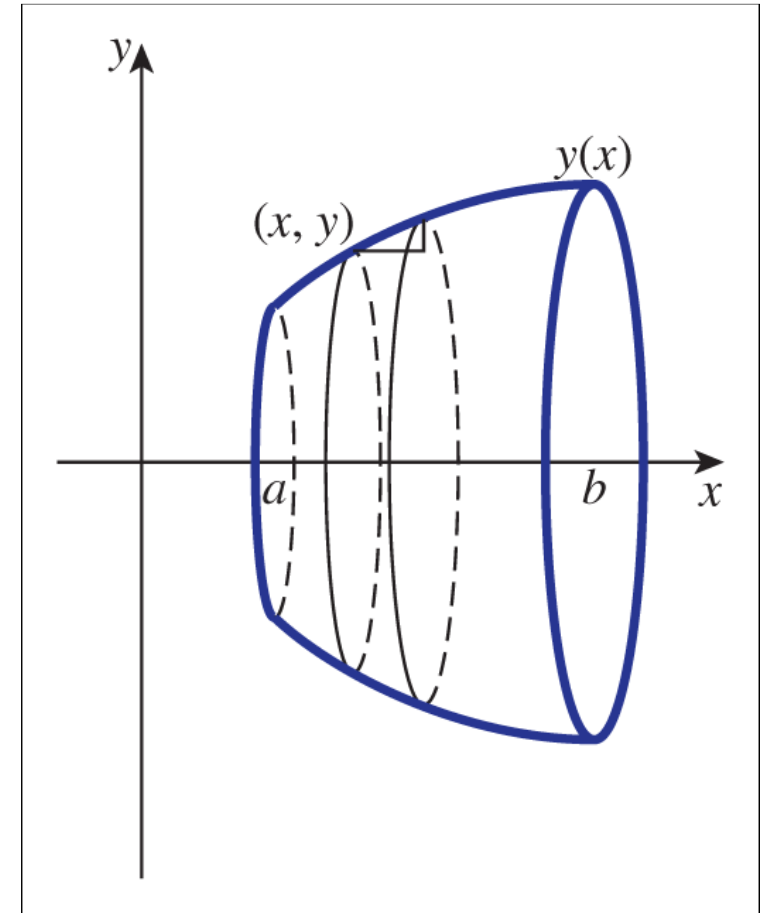
$$\text{length of curve} = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Area of a Surface of Revolution

$$\text{area of surface} = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

We can prove this formula by:

- 1 selecting a small disc of the solid of revolution;
- 2 calculating the length of its circumference;
- 3 approximating the surface area of this disc using Pythagora's for the length of the curve;
- 4 summing all contributions to the surface area between $x = a$ and $x = b$, and letting $\delta x \rightarrow 0$.



Area of a Surface of Revolution

10. The line $y = x$ between $x = 0$ and $x = 1$ is rotated around the x axis.

(a) Find the area of the surface generated.

⑩ $y = x, \quad a = 0$
 $b = 1$

$$a) \text{ Area} = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= \int_0^1 2\pi x \sqrt{1+1} dx = 2\pi\sqrt{2} \int_0^1 x dx$$

$$= 2\pi\sqrt{2} \left[\frac{x^2}{2} \right]_0^1 = \pi\sqrt{2}$$

L6. Mean and RMS values

$$\text{average value} = h = \frac{1}{b-a} \int_a^b f(t) \, dt$$

$$\text{r.m.s value} = \sqrt{\frac{1}{b-a} \int_a^b [f(t)]^2 \, dt}$$

L7. Complex Numbers

Representations

Rectangular/
Cartesian $z = a + ib$

Polar $z = r \angle \theta \rightarrow z = r(\cos \theta + i \sin \theta)$

Exponential $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$

Polar $\longleftrightarrow$ Cartesians

$$\theta = \arg(z) = \tan^{-1} \left(\frac{b}{a} \right)$$

$$r = |z| = \sqrt{a^2 + b^2}$$

Multiplication / Division

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle (\theta_1 - \theta_2)$$

$$z_1 z_2 = r_1 r_2 \angle (\theta_1 + \theta_2),$$

Euler's relations

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

L7. Complex Numbers

Polar to Cartesian

$$z = 4\angle\frac{\pi}{3}$$

(angle / polar form)

$$= 4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

(definition)

$$= 4\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$$

(evaluate trig)

$$= 2 + 2\sqrt{3}i$$

(Cartesian form)

Cartesian to Polar

$$z = 2 + 2\sqrt{3}i$$

(Cartesian form)

$$r = \sqrt{2^2 + (2\sqrt{3})^2}$$

(modulus)

$$= \sqrt{4 + 12} = \sqrt{16} = 4$$

$$\theta = \tan^{-1}\left(\frac{2\sqrt{3}}{2}\right)$$

(argument)

$$= \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

Form	Expression		
Cartesian	$2 + 2\sqrt{3}i$	Trigonometric	$4(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$
Polar	$4\angle\frac{\pi}{3}$	Exponential	$4e^{i\pi/3}$

L7. Complex Numbers – De Moivre's Theorem

$$z^n = r^n \angle n\theta$$

$$z^n = (r(\cos \theta + i \sin \theta))^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta)$$

If $z = \cos \theta + i \sin \theta$

$$z^n + \frac{1}{z^n} = 2 \cos n\theta$$

$$z^n - \frac{1}{z^n} = 2i \sin n\theta$$

L7. Complex Numbers – Solving Equations

To solve a complex number equations such as $z^3 = 4$

- 1 Write down the RHS of the equation in polar/ exponential form **with a general argument**.

$$4 = 4e^{i(0+2k\pi)}$$

- 2 Equate the LHS to the RHS

$$z^3 = 4e^{i(2k\pi)}$$

- 3 Take the root of both sides of the equations (depending on the order of the equation)

$$z = \sqrt[3]{4e^{i(2k\pi)}} = 4^{1/3} e^{i2k\pi/3}$$

- 4 Cycle through values of k , starting from 0, until you have found all roots (i.e. $k = 0 \rightarrow n - 1$)

$$k = 0 \Rightarrow z_0 = 4^{1/3}, \quad k = 1 \Rightarrow z_1 = 4^{1/3} e^{i2\pi/3}, \quad k = 2 \Rightarrow z_2 = 4^{1/3} e^{i4\pi/3}$$

- 5 Plot the roots on the Argand Diagram and check they are evenly spaced (and that you have n of them!)

L7. Complex Numbers – Solving Equations

2) Find the roots of $(z-3i)^4 = 3i$

Rewrite $3i = 3e^{i\left(\frac{\pi}{2} + 2k\pi\right)}$

$$(z-3i)^4 = 3e^{i\left(\frac{\pi}{2} + 2k\pi\right)}$$

$$z-3i = 3^{\frac{1}{4}} e^{i\left(\frac{\pi}{8} + k\frac{\pi}{2}\right)}$$

$$k=0 \Rightarrow z = 3^{\frac{1}{4}} e^{i\frac{\pi}{8}} + 3i$$

$$k=1 \Rightarrow z = 3^{\frac{1}{4}} e^{i\left(\frac{\pi}{8} + \frac{\pi}{2}\right)} + 3i$$

$$k=2 \Rightarrow z = 3^{\frac{1}{4}} e^{i\left(\frac{\pi}{8} + \pi\right)} + 3i$$

$$k=3 \Rightarrow z = 3^{\frac{1}{4}} e^{i\left(\frac{\pi}{8} + \frac{3}{2}\pi\right)} + 3i$$

L8. Sequences & Series

The sum of the first n positive integers is given by

$$\sum_{k=1}^n k = 1 + 2 + 3 + \cdots + n = \frac{n}{2}(n+1)$$

The sum of the squares of the first n positive integers is given by

$$\sum_{k=1}^n k^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n}{6}(n+1)(2n+1)$$

The sum of the cubes of the first n positive integers is given by

$$\sum_{k=1}^n k^3 = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

L8. Sequences & Series

The **k -th term** of an arithmetic sequence is given by

$$x_k = a + (k - 1)d, \text{ where } a \text{ is the first term and } d \text{ is the common difference.}$$

The **sum** of the first n terms of an arithmetic sequence is

$$S_n = \frac{n}{2} (2a + (n - 1)d).$$

The **k -th term** of a geometric sequence is given by

$$x_k = ar^{(k-1)}, \text{ where } a \text{ is the first term and } r \text{ is the common ratio.}$$

The **sum** of the first n terms of a geometric sequence is

$$S_n = \frac{a(1 - r^n)}{1 - r} \quad (\text{valid only if } r \neq 1)$$

L8. Sequences & Series - Expansions

We want to expand $(a + b)^n$

When n is small: Pascal's Triangle

			1			
		1		1		
	1		2		1	
1		3		3		1
1	4	6	4	1		
1	5	10	10	5	1	

$$\longrightarrow (a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

When n is large: Binomial Theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k,$$

where n is an integer, a , b are real numbers, and

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}, \quad n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 2 \cdot 1.$$

L8. Sequences & Series - Expansions

We want to expand $(a + b)^n$

When n is small: Pascal's Triangle

		1			
	1		1		
	1	2	1		
1	3	3	1		
1	4	6	4	1	
1	5	10	10	5	1

$$\longrightarrow (a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

When n is not a positive integer and x is within $(-1, 1)$

$$(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \dots$$

L8. Sequences & Series – Maclaurin Series

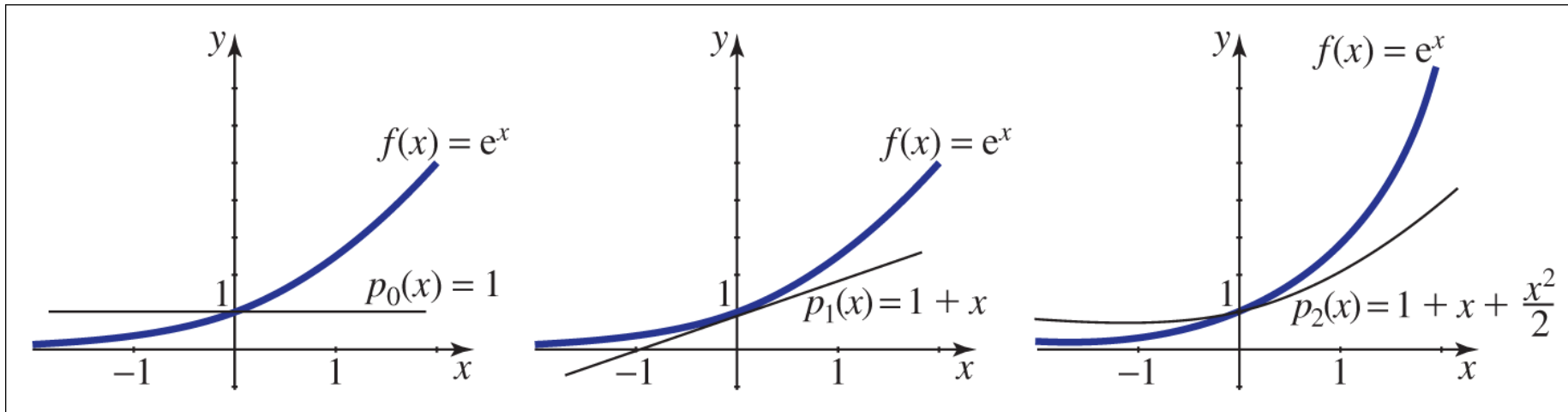
Power series expansion to approximate of any n-derivable function around x=0

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(0) x^n = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots,$$

where $f^{(n)}(0)$ denotes the n -th derivative of f at $x = 0$.

Visual Representation

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$



L8. Sequences & Series – Power Series Expansions

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \text{ for all } x$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \text{ for all } x$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \text{ for all } x$$

$$\log_e(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \text{ for } -1 < x \leq 1$$

$$(1 + x)^p = 1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \cdots \text{ for } -1 < x < 1$$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \cdots \text{ for } -\frac{\pi}{2} < x < \frac{\pi}{2}$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots \text{ for all } x$$

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots \text{ for all } x$$

L8. Sequences & Series – Taylor Series

Power series expansion to approximate of any n-derivable function around $x=x_0$

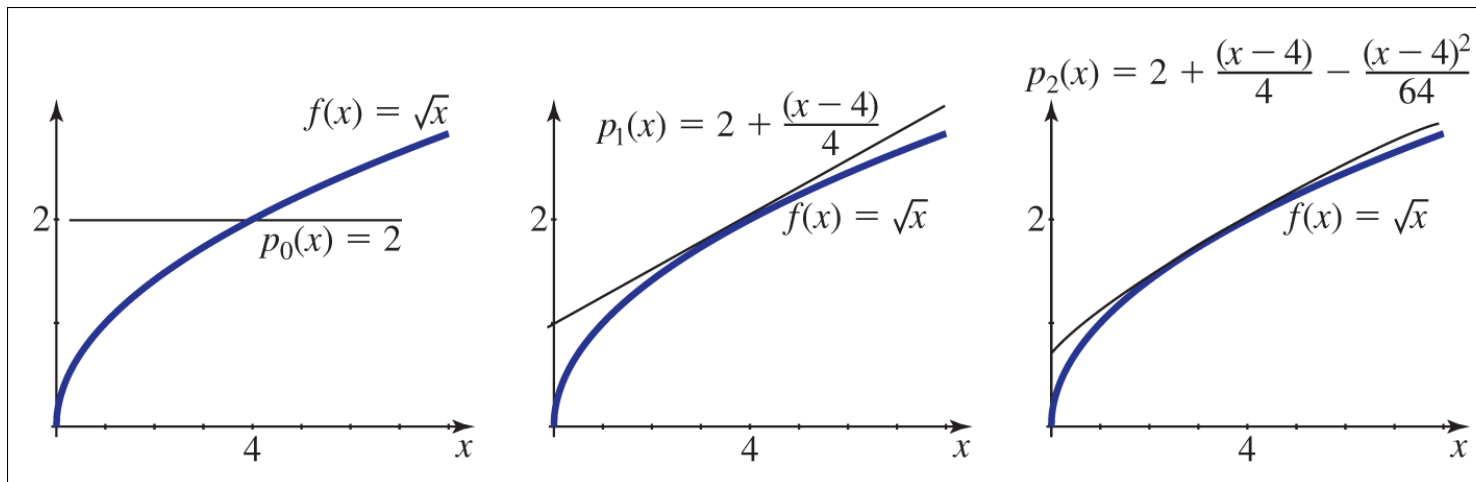
$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(x_0)(x - x_0)^n$$

where $f^{(n)}(x_0)$ denotes the n -th derivative of f at $x = x_0$.

$$= f(x_0) + (x - x_0)f'(x_0) + \frac{(x - x_0)^2}{2!}f''(x_0) + \frac{(x - x_0)^3}{3!}f'''(x_0) + \dots,$$

Visual Representation

$$\sqrt{x} = 2 + \frac{1}{4}(x - 4) - \frac{1}{64}(x - 4)^2 + \frac{1}{512}(x - 4)^3 + \dots$$



L8. Sequences & Series – Taylor Series

7. Find the Taylor series for $x + e^x$ about $x = 1$.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(1)}{n!} (x - 1)^n$$

Compute the derivatives & evaluate at $x=1$

$$f(x) = x + e^x$$

$$f(1) = 1 + e$$

$$f'(x) = 1 + e^x$$

$$f'(1) = 1 + e$$

$$f''(x) = e^x$$

$$f''(1) = e$$

$$f^{(n)}(x) = e^x \quad \text{for } n \geq 2$$

$$f^{(n)}(1) = e \quad (n \geq 2)$$

Substitute

$$x + e^x = (1 + e) + (1 + e)(x - 1) + \frac{e}{2!}(x - 1)^2 + \frac{e}{3!}(x - 1)^3 + \dots$$

Compact form

$$x + e^x = (1 + e) + (1 + e)(x - 1) + \sum_{n=2}^{\infty} \frac{e}{n!} (x - 1)^n$$

L9. Differential Equations

General form

$$\frac{dy(t)}{dt} = f(t, y(t))$$

Ordinary Diff. Eq. (ODE)

$$\frac{d^n y}{dx^n} + f\left(x, y, \frac{dy}{dx}, \dots, \frac{d^{n-1}y}{dx^{n-1}}\right) = 0$$

y : dependent variable
x : independent variable

Linear ODE

$$\sum_{j=0}^n p_j(x) \frac{d^j y}{dx^j} + g(x) = 0, \quad g(x) = 0 : \text{Homogeneous}$$

First order linear ODE

$$\frac{dy}{dx} + f(x, y) = 0$$

L9. Differential Equations – Separation of Variables

For separable, nonlinear, equations

The solution of the equation

$$\frac{dy}{dx} = f(x)g(y)$$

is found from

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

Example: Bernoulli

$$\frac{dy}{dx} + p(x)y = f(x)y^r,$$

We can transform it into a separable equation by taking $y(x) = u(x)y_1(x)$, with $y_1(x)$ a nontrivial solution of the homogeneous equation ($f(x)=0$), which needs to be found.

Substitution of $y(x) = u(x)y_1(x)$ into the original equation yields a separable equation:

$$\frac{du}{dx} = u(x)^r f(x)(y_1(x))^{r-1}.$$

L9. Differential Equations – Bernoulli

④ a) $y' - y = xy^2$

Homogeneous equation: $y' - y = 0$, $y_1 = e^x$

We look for solutions in the form $y = ue^x$. Thus,

$$u'e^x + \cancel{ue^x} - \cancel{ue^x} = xu^2e^{2x}, \quad u' = xu^2e^x$$

Separating variables yields

$$\frac{du}{u^2} = xe^x dx$$

and integrating yields

$$-\frac{1}{u} = \int xe^x dx \stackrel{\substack{\uparrow \\ \text{int. by parts}}}{=} xe^x - \int e^x dx + c = xe^x - e^x + c = e^x(x-1) + c$$

$$u = -\frac{1}{(x-1)e^x + c}$$

$$y = ue^x = -\frac{1}{x-1+ce^{-x}}$$

L9. Differential Equations – Integrating Factor

All first-order linear ODEs can be written in standard form as

$$\frac{dy}{dx} + P(x)y = Q(x)$$

- 1 Multiply by a function $\mu(x)$ known as **integrating factor**; $\mu = e^{\int P(x)dx}$

The equation simplify as $\frac{d}{dx}(\mu y) = \mu Q(x)$

- 2 Integrate both sides;
- 3 Divide by $\mu(x)$.

L9. Differential Equations – Integrating Factor

A. Rewrite in standard form

$$\frac{dy}{dx} + 2y = e^{-2x} \quad \begin{array}{l} P(x) = 2 \\ Q(x) = e^{-2x} \end{array}$$

B. Choose the integrating factor μ .

$$\mu = e^{\int P(x) dx} = e^{\int 2 dx} = e^{2x}$$

C. Multiply by μ

$$e^{2x} \frac{dy}{dx} + 2e^{2x} y = e^{2x} \cdot e^{-2x}$$

$$\frac{d}{dx} (e^{2x} y) = 1$$

Note that this is because we have defined $\mu(x)$ as the function for which:

$$\frac{d}{dx} (\mu(x) \cdot y) = \mu(x) Q(x)$$

L9. Differential Equations – Integrating Factor

D. Integrate : $\int \frac{d}{dx} (e^{2x} y) dx = \int dx$

$$e^{2x} y = x + c$$

E. Divide by the integrating factor $\mu(x) = e^{2x}$

$$e^{-2x} \cdot e^{2x} \cdot y = (x + c) \cdot e^{-2x}$$

$$y = (x + c) e^{-2x} \rightarrow \text{general solution}$$

L10. Partial Derivatives

The (first) partial derivative wrt x of a function $z = f(x, y)$ is denoted by $\frac{\partial z}{\partial x}$ and is calculated by differentiating the function wrt x and treating y as though it were a constant.

$$8) z = \sin(3x - y^2)$$

$$\text{chain rule} \rightarrow f(x) = g(h(x)) \quad f'(x) = g'(h(x)) \cdot h'(x)$$

outer function $\sin(\cdot)$

inner function $3x - y^2$

$$\frac{\partial z}{\partial x} = \underbrace{\cos(3x - y^2)}_{\text{outer}} \cdot \underbrace{3}_{\text{inner}}$$

$$\frac{\partial z}{\partial y} = \underbrace{\cos(3x - y^2)}_{\text{outer}} \cdot \underbrace{(-2y)}_{\text{inner}} = -2y \cos(3x - y^2)$$

L10. Partial Derivatives

The (first) partial derivative wrt x of a function $z = f(x, y)$ is denoted by $\frac{\partial z}{\partial x}$ and is calculated by differentiating the function wrt x and treating y as though it were a constant.

$$c. \quad z = x \ln(xy)$$

$$\frac{\partial z}{\partial x} = \ln(xy) + \frac{1}{xy} \cdot y \cdot x = \ln(xy) + 1$$

$$\frac{\partial z}{\partial y} = \frac{x \cdot x}{xy} = \frac{x}{y}$$

L10. Partial Derivatives – Second order

differentiating $\frac{\partial z}{\partial x}$ wrt x produces $\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial^2 z}{\partial x^2}$

differentiating $\frac{\partial z}{\partial x}$ wrt y produces $\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial^2 z}{\partial y \partial x}$

differentiating $\frac{\partial z}{\partial y}$ wrt x produces $\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial^2 z}{\partial x \partial y}$

Same

differentiating $\frac{\partial z}{\partial y}$ wrt y produces $\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial^2 z}{\partial y^2}$

L10. Partial Derivatives – Stationary Values

Stationary Points

$$\frac{\partial f}{\partial x} = 0$$

$$\frac{\partial f}{\partial y} = 0$$

Type of Stationary Points

$$\frac{\partial^2 f}{\partial^2 x} \frac{\partial^2 f}{\partial^2 y} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 < 0 \quad \Rightarrow \text{saddle point}$$

$$\frac{\partial^2 f}{\partial^2 x} \frac{\partial^2 f}{\partial^2 y} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 > 0 \quad \text{and} \quad \frac{\partial^2 f}{\partial^2 x} > 0 \quad \Rightarrow \text{minimum point}$$

$$\frac{\partial^2 f}{\partial^2 x} \frac{\partial^2 f}{\partial^2 y} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 > 0 \quad \text{and} \quad \frac{\partial^2 f}{\partial^2 x} < 0 \quad \Rightarrow \text{maximum point}$$

L10. Partial Derivatives – Stationary Values

Locate and identify the nature of the stationary values of $f(x, y) = x^2 + xy - 7y$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} (2x + y) = 2$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} (x - 7) = 0$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} (x - 7) = 1$$

$$\Rightarrow \frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial^2 f}{\partial y^2} = 0$$

$$\frac{\partial^2 f}{\partial x \partial y} = 1$$

$$\frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} < \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 \Rightarrow \text{SADDLE POINT}$$

L10. Multiple Integrals

Double integrals

$$\int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x, y) dx dy$$

- $f(x, y)$ is first integrated wrt x (holding y constant) between the limits of $x = x_1$ and $x = x_2$.
- the result is the integrated wrt y between the limits $y = y_1$ and $y = y_2$.

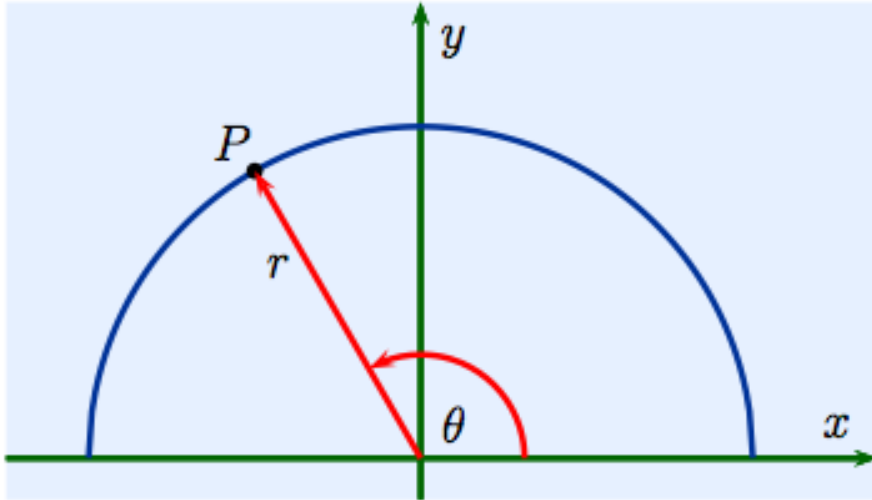
With non-constant limits

$$I = \int_{x=a}^{x=b} dx \int_{y=y_1(x)}^{y=y_2(x)} f(x, y) dy$$

Integration over area in 2D (constant limits)

$$A = \int_{x=a}^{x=b} \int_{y=c}^{y=d} dy dx$$

L10. Multiple Integrals – Polar coordinates



$$x = r \cos \theta \quad y = r \sin \theta$$

$$\tan \theta = \frac{y}{x} = \left(\frac{r \sin \theta}{r \cos \theta} \right)$$

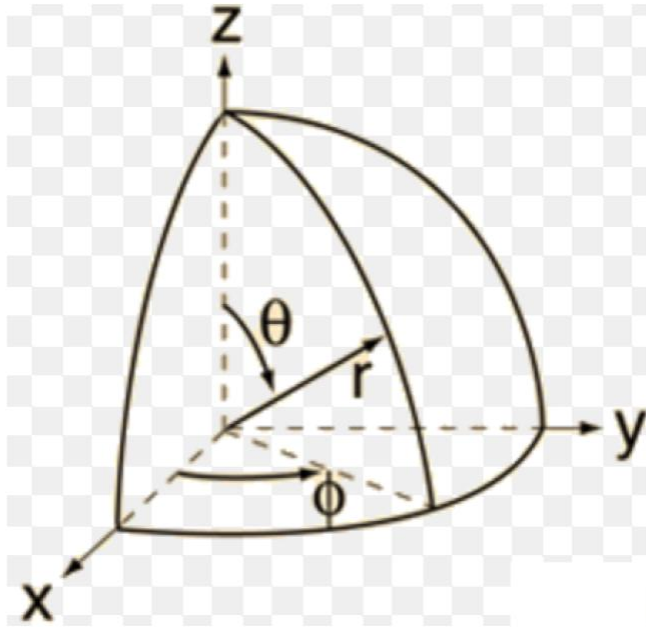
$$r = \sqrt{x^2 + y^2}$$

$$I = \int \int_R f(r, \theta) r dr d\theta \quad \text{Note that } dA = dxdy = r d\theta dr$$

Limits of integration

$$r : 0 \rightarrow \infty \quad \theta : 0 \rightarrow 2\pi$$

L10. Multiple Integrals – Spheric coordinates



Spherical $\rightarrow$ Cartesian

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

Cartesian $\rightarrow$ Spherical

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1} \left(\frac{z}{r} \right)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$I = \int \int \int_R f(r, \theta, \psi) r^2 \sin \theta dr d\theta d\psi \quad \text{Note that} \quad dV = dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

Limits of integration

$$r : 0 \rightarrow \infty$$

$$\theta : 0 \rightarrow \pi$$

$$\psi : 0 \rightarrow 2\pi$$